PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Smart AI System for Short Video Ad Creation

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Smart AI System for Short Video Ad Creation"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **"Gurpreet Singh"**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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1. Introduction

Short video advertisements have become one of the most effective tools in digital marketing, especially with the rise of platforms like Instagram Reels, YouTube Shorts, and TikTok. These platforms attract a large audience and enable businesses to promote their products in an engaging and visually appealing way. However, creating such ads manually requires significant time, creativity, professional photography, and technical expertise in editing. This makes it difficult for small businesses and individuals to produce high-quality advertisement content consistently.

To address this challenge, artificial intelligence can be leveraged to automate and enhance the ad creation process. This project introduces an AI-based system that allows users to upload a product image and a model image, which are then intelligently combined to generate realistic advertisement visuals. The system further converts these visuals into short video ads, integrating automation with intelligent content generation. By reducing manual effort and eliminating the need for photoshoots, the solution makes ad creation faster, cost-effective, and accessible to a wider audience.

2. Methodology

The methodology of this project focuses on developing an automated pipeline for generating short video advertisements using artificial intelligence. The process begins with user input, where the user uploads a product image, a model image, and optionally provides prompts such as style, theme, or target audience.

The system then uses AI-based image generation techniques to seamlessly combine the product and model images into a single realistic advertisement visual. After generating the visual, the system creates an engaging script based on the provided input using natural language processing techniques.

Next, the generated visual is transformed into a short video advertisement by adding animations, transitions, and effects. Voice-over narration and background music are automatically integrated to enhance the quality and engagement of the video. Finally, all components are compiled into a complete, ready-to-use short video advertisement suitable for modern digital marketing platforms.

3. Tools and Technologies

3.1 Tools

1. **Visual Studio Code**
   Code editor for development and debugging
2. **Postman**
   Used for testing APIs and backend services
3. **Git & GitHub**
   Version control and project collaboration

3.2 Technologies Used

- **PERN Stack (Core Technology)**

The project is built using PostgreSQL, Express.js, React, and Node.js for full-stack development.

- **Neon Database (PostgreSQL - Cloud)**

Neon is used as a serverless PostgreSQL database to store user data, ad content, and video details with high scalability.

- **Express.js (Backend Framework)**

Handles API requests and manages communication between frontend and database efficiently.

- **React (Frontend)**

Builds a dynamic and responsive user interface using component-based architecture.

- **Node.js (Runtime Environment)**

Executes server-side logic and supports scalable backend operations.

- **Clerk (Authentication)**

Provides secure user authentication, including login, signup, and session management.

- **Gemini API (AI Integration)**

  Used for generating ad scripts and intelligent content using AI capabilities.

- **Data Sanitization**

  Ensures user inputs are cleaned and validated to prevent errors and security issues.

- **REST APIs**

  Enable smooth communication between frontend, backend, and external services.

4. Implementation

1. Frontend Development (React)

   The user interface is built using React, where users can enter product details, target audience, and preferences. It provides a smooth and interactive experience.

2. Backend Processing (Node.js & Express.js)

   The backend handles user requests, processes input data, and manages communication between frontend, database, and AI services.

3. Database Management (Neon PostgreSQL)

   A cloud-based PostgreSQL database (Neon) is used to store user information, generated scripts, and video-related data securely and efficiently.

4. AI Integration (Gemini API)

   The system uses Gemini API to generate creative and engaging ad scripts based on user input, making the content intelligent and automated.

5. Media Processing (FFmpeg)
   FFmpeg is used to process and combine images, video clips, voiceovers, and background music into a final video output.

6. Authentication System (Clerk)
   Clerk is implemented for secure user authentication, including login, signup, and session management.

7. Data Sanitization & Validation
   User inputs are cleaned and validated to prevent errors, ensure data consistency, and improve system security.

8. Final Output Generation
   All components are integrated to generate a complete short video advertisement that is ready to publish on social media platforms.

5.Workflow

The workflow of the system begins when the user uploads a product image and a model image, along with optional inputs such as product description, target audience, and preferred tone. These inputs form the foundation of the advertisement creation process.

The system then uses AI-based image generation techniques to seamlessly combine the product and model images into a single realistic advertisement visual. Based on the provided inputs, the AI also generates an engaging and relevant script.

Next, the generated visual is transformed into a short video advertisement by adding animations, transitions, and effects. The system further enhances the video by incorporating AI-generated voice-over and suitable background music that aligns with the tone of the advertisement.

Finally, all components are integrated to produce a complete, ready-to-publish short video ad optimized for social media platforms.

6. **Advantages**

I. Automation of Entire Process
The system automates end-to-end ad creation from script to final video. This reduces dependency on multiple tools and manual steps.

II. Faster Content Creation
Ads can be generated in minutes instead of hours or days. This helps businesses respond quickly to trends and market demands.

III. User-Friendly Interface
The platform is designed to be simple and easy to use. Even non-technical users can create ads with minimal effort.

IV. Consistency in Branding
AI helps maintain a consistent tone, style, and quality across all advertisements. This strengthens brand identity.

V. Customizable Content
Users can modify inputs like tone, audience, and product details. This allows flexible and tailored ad creation.

VI. Reduced Human Error
Automation minimizes mistakes that can occur during manual editing. This improves reliability and output quality.

VII. Accessible Anytime, Anywhere

Being cloud-based, the system can be accessed from anywhere. This increases convenience and flexibility for users.

VIII. Improved Productivity

Businesses can create multiple ads in less time. This boosts overall marketing efficiency and output.

7. Use cases

The AI Short Video Ads Generator can be applied across various domains in digital marketing and content creation. It is especially useful for e-commerce businesses, where it can generate realistic product advertisements by combining product and model images without the need for physical photoshoots. The system is highly effective for social media marketing, as it produces short, engaging video ads optimized for platforms like Instagram, YouTube, and TikTok.

Startups and small businesses can use this system to create professional-quality ads at a low cost, reducing dependency on designers and marketing teams. Additionally, it supports personalized advertising by allowing users to customize inputs, enabling targeted and audience-specific content creation.

8. Results

The implementation of the AI Short Video Ads Generator demonstrates that artificial intelligence can significantly improve both the efficiency and creativity of advertisement production. The system successfully automates the process of combining product and model images to generate realistic advertisement visuals and converts them into short video ads.

It reduces the time, effort, and cost required compared to traditional methods while maintaining high-quality output. The integration of script generation, AI-based visual creation, and automated

video production ensure consistency and engagement. Overall, the results show that the system is effective, scalable, and capable of simplifying modern ad creation.

9. Future Scope

The future scope of this project includes enhancing the system with advanced personalization features that allow advertisements to be tailored according to user behavior, preferences, and target audience. By leveraging more sophisticated AI models, the system can generate highly customized and engaging content, making ads more relevant and effective for different users and markets.

Another important improvement is the integration of real-time ad generation and multilingual support. This will enable the system to create and update ads instantly based on current trends and user inputs while also supporting multiple languages to reach a global audience. Additionally, direct integration with social media platforms can allow seamless publishing and better optimization of ads.

Furthermore, the system can be expanded by incorporating features like voice cloning, AI-based avatars, and detailed analytics for performance tracking. These enhancements will not only improve user experience but also provide valuable insights into audience engagement, helping businesses create more impactful and data-driven advertisements.

10. Conclusion

In conclusion, the AI Short Video Ads Generator demonstrates how artificial intelligence can simplify and enhance the process of creating short video advertisements. By combining product and model images using AI and automating key tasks such as script generation and video creation, the system significantly reduces time, cost, and manual effort.

It provides an efficient and accessible solution for businesses and individuals to create high-quality, engaging advertisements without requiring professional skills or photoshoots. Overall, the project highlights the transformative potential of AI in digital marketing through automation, creativity, and scalability.